

Program : Diploma in Robotic Process Automation	
Course Code : 3304	Course Title: Introduction to Software Engineering
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4(L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide the fundamental concepts of programming methodology and software engineering.
- To introduce various software process /lifecycle models.
- To compare and contrast structured vs object oriented modeling.
- To understand various testing strategies.
- To understand various principles of software project management.
- Identify evolving trends in software Engineering with DevOps.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge on problem solving and programming concepts.		Introduction to computation and programming.	2

Course Outcomes:

On completion of the course student will be able to:

Con	Description	Duration (Hours)	Cognitive Level
CO1	Explain about software life cycle phases and various models used in Software Development Process.	15	Understanding
CO2	Illustrate requirements, specifications, analysis and design approaches for the software development process.	12	Understanding

CO3	Discuss about software implementation and testing strategy for the software system.	15	Understanding
CO4	Illustrate the planning and management of software projects.	16	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

	Description	Duration (Hours)	Cognitive Level
CO1	Explain about software life cycle phases and various models used in Software Development Process.		
M1.01	Discuss about programming Methodologies	1	Understanding
M1.02	Understand structured approach.	1	Understanding
M1.03	Explain about modular approach-top down and bottom up.	1	Understanding
M1.04	Explain about Object oriented approach-Class, Object, inheritance, Polymorphism.	2	Understanding
M1.05	Discuss about the software crisis and emergence of software engineering .	1	Understanding.

M1.06	Explain about various phases of software development .	1	Understanding
M1.07	Discuss about Software lifecycle models.	2	Understanding
M1.08	Explain about Classical waterfall, Iterative, prototyping, incremental and Spiral .	4	Understanding
M1.09	Explain about Extreme programming and Agile processes.	2	Understand

Contents:

Programming Methodologies - Structured approach, Modular programming, Objectoriented approach-Class, Object, inheritance, Polymorphism, **Introduction to software engineering-software** crisis, emergence of software engineering, Phases of software development. **Software life cycle models** –Classical waterfall, Iterative, prototyping, incremental and Spiral. Extreme programming and Agile processes.

CO2:	Illustrate requirements, specifications, analysis and design approaches for the software development process.		
M2.01	Explain about requirements analysis	1	Understanding.
M2.02	Discuss the importance of requirements specification.	1	Understanding.
M2.03	Illustrate SRS document preparation and its importance.	2	Understanding.
M2.04	Explain about software design concepts-Cohesion and coupling	1	Understanding.
M2.05	Illustrate the concept of function Oriented Design and its Complexity Metrics(DFD)	3	Understanding.
M2.06	Illustrate the concept of Object Oriented Design and its Complexity Metrics.(UML)	4	Understanding.
	Series Test – I	1	

Contents:

Requirements Analysis and specification ,SRS-characteristics and structure. Software Design- Design heuristics-. Cohesion and Coupling ,Function Oriented Design and its Complexity Metrics ,Object Oriented Design and its Complexity Metrics.

CO3:	Discuss about software implementation and testing strategy for the software system.		
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M3.01	Understand programming principles and coding guidelines.	2	Understanding
M3.02	Understand the method of incrementally developing code.	2	Understanding
M3.03	Discuss about software testing-test case and test plan	2	Understanding
M3.04	Explain about white box and black box testing.	5	Understanding
M3.05	Understand the concept of unit testing and integration testing.	3	Understanding
M3.06	Explain about system testing.	1	Understanding

Contents:

Software Implementation and Testing-Programming principles and coding guidelines, method of incrementally developing code, **Software Testing**-Test case and Test plan ,white box testing, black box testing, Unit testing, integration testing, System testing.

CO4:	Illustrate the planning and management of software projects.		
M4.01	Explain software project management and its phases.	1	Understanding
M4.02	Discuss about Software planning, software organizing, Staffing, Directing and controlling	5	Understanding
M4.03	Describe the responsibilities of project manager	1	Understanding
M4.04	Write brief description about project estimation techniques for project planning	1	Understanding
M4.05	Understand the concept of Software Configuration Management(SCM)	2	Understanding.
M4.06	Differentiate Quality Standards-ISO and CMM.	2	Understanding
M4.07	Understand Devops-Introduction-Definition, Why Devops? Goals, Benefits	2	Understanding.
M4.08	Discuss about Devops tools and relationship to agile and devops (continuous delivery)	2	Understanding
	Series Test – II	1	

Contents:

Software Project Management:Brief study of various phases of project management-Planning,Organizing,staffing ,directing and controlling.**Software project cost estimation-**COCOMO model,Software Configuration Management(SCM),**Quality Standards-**ISO and CMM. **Devops-**Introduction-Definition, Why Devops?, Goals, Benefits, Relationship to Agile and Devops (continuous delivery), Devop Tools.

Text / Reference

T/R	Book Title/Author
T1	Fundamentals of software engineering, Fourth Edition, Rajib Mall.PHI
T2	Software Engineering, A Precise Approach: Pankaj Jalote, Wiley India-2010
T3	Software Project Management : Saikat Dutt /S. Chandramouli, Pearson-Second Edition
R1	Software Engineering : Ian Sommerville, Pearson,Ninth Edition
R2	Software Engineering a practitioner's approach – Roger S Pressman,Seventh Edition
R3	Project Management Absolute Beginner's Guide : Greg Horine , Pearson, Second Edition

Online Resources

Sl.No	Website Link
1	https://en.wikipedia.org/wiki/DevOps
2	http://www.informationweek.com/devops/agile-vs-devops-10-ways-theyredifferent/d/d-id/1326121